

4.7 Detecting multiple features with branches

Settings

Beta features

PythonTutor: Multiple distinct if statements.

```
→ user_age = 26 # Hardcoded for this tool. Could replace with "i

# Note that more than one "if" statement can execute
if user_age < 16:
    print("Enjoy your early years.")

if user_age > 15:
    print("You are old enough to drive.")

if user_age > 17:
    print("You are old enough to vote.")

if user_age > 24:
    print("Most car rental companies will rent to you.")

if user_age > 34:
    print("You can run for president.")
```

→ line that just executed

→ next line to execute



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Step 1 of 9

Program output

CHALLENGE ACTIVITY

4.7.1: Enter the output for the multiple if-else branches.

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Start

Type the program's output

```
num_items = 6

if num_items < 1:
    print("b")
elif num_items < 7:
    print("e")
else:
    print("g")

print("m")
```

e
m

1

2

3

4

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nested if-else

A branch's statements can include any valid statements, including which are known as nested if-else statements.

PythonTutor: Nested if-else

```
➔ user_choice = 2 # Hardcoded values for this tool. Cou
num_items = 5

if user_choice == 1:
    print("user_choice is 1")
elif user_choice == 2:
    if num_items < 0:
        print("user_choice is 2 and num_items < 0")
    else:
        print("user_choice is 2 and num_items >= 0")
else:
    print("user_choice is neither 1 or 2")
```

➡ line that just executed

➔ next line to execute

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Step 1 of 6

Program output

CHALLENGE ACTIVITY

4.7.2: Detecting multiple features with branches.

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Start

health_points is read from input. Write multiple if statements:

- If health_points is greater than 22, then output "I can run a mile."
- If health_points is greater than 39, then output "I can run a marathon."
- If health_points is less than or equal to 3, then output "I am tired."

[Click here for examples](#) ▼

```
1 health_points = int(input())
2
3 """ Your code goes here """
4
```

1

2

Check**Next level**

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